ViDA-UGC: Detailed Image Quality Analysis via Visual Distortion Assessment for User-Generated Content

Wenjie Liao, Jieyu Yuan, Yifang Xu, Chunle Guo *Member, IEEE*, Jihong Li,
Zilong Zhang, Jiachen Fu, Haotian Fan, and Chongyi Li, *Senior Member, IEEE*

***Abstract*—Recent multimodal large language models (MLLMs) have shifted image quality assessment (IQA) from black-box quality scoring toward explainable quality analysis. However, existing datasets and benchmarks are not specifically designed for user-generated content (UGC) distortions, which arise from real-world capture, processing, and sharing pipelines. They also lack a unified framework for detailed IQA. In this study, we propose detailed IQA for UGC images, formulated around three core abilities: *Grounding* (A1), *Perception* (A2), and *Description* (A3). To support these abilities, we establish the first large-scale Visual Distortion Assessment Instruction Tuning Dataset for UGC images, named ViDA-UGC, containing 11,534 images, 36K distortion bounding boxes, and 534K instruction tuning data. This dataset is built through a distortion-oriented pipeline combining human subject annotation with a Grounding-Perception-Guided Controlled Chain-of-Thought (GPGC CoT) prompting, which guides GPT-4o to progressively identify, analyze, and reason about UGC distortions along an explicit distortion evidence chain. We further select and verify a subset of ViDA-UGC with a professional team to build ViDA-UGC-Bench, the first UGC distortion assessment benchmark covering all three abilities. Experimental results show that ViDA-UGC fine-tuning consistently enhances grounding, perception, and description across Qwen-VL and InternVL series, while GPGC CoT prompting further improves quality descriptions without fine-tuning, with the best-tuned models even surpassing GPT-4o. ViDA-UGC-Bench further reveals that current MLLMs remain limited in distortion assessment for detailed IQA. The dataset and code will be released publicly.**

***Index Terms*—Multi-modal large language models, image quality assessment, visual distortion, distortion grounding, instruction tuning.**

I. Introduction

Image Quality Assessment (IQA) seeks to evaluate image perceptual quality in a way that is consistent with the human visual system. With the rapid growth of digital visual content, IQA has become increasingly important in media transmission, user-generated photography, smartphone imaging, and many other real-world applications. Attributed to the emergence of multimodal large language models (MLLMs) [1]–[4], this field is now shifting from black-box quality scoring [5]–[8] toward explainable image quality assessment [9]–[12]. Instead of producing only a scalar score, explainable IQA leverages MLLMs' cross-modal capabilities to deliver human-readable quality analysis that can guide quality control, camera systems, and visual enhancement.

In this work, we push explainable IQA one step further toward **detailed IQA**, which is not limited to quality judgment but requires explaining *what* degrades image quality and *where* the relevant evidence appears, *how* severe the degradation is, and *why* it affects the final judgment. For user-generated content (UGC) images, this goal naturally leads to visual distortion assessment, since human quality analysis usually follows an interpretable path from distortions to judgment: first identifying distorted evidence*what and where*), then perceiving its low-level attributes*how*) and finally organizing these observations into explicit quality analysis*why*). Therefore, we formulate detailed IQA for UGC around three core abilities along this path: ***Grounding (A1)*** localizes quality-relevant distortion evidence itself, anchoring later reasoning to degradations that directly support the judgment; ***Perception (A2)*** reads fine-grained distortion attributes, such as type, position, severity, impact, and significance, as the core evidence for quality analysis; and ***Description (A3)*** organizes grounded and perceived evidence into explicit reasoning, for both overall quality analysis and individual distortion assessment.

However, existing explainable IQA datasets and benchmarks rarely organize these three abilities under a unified framework. Most perception- and description-oriented instruction tuning datasets [10]–[14] strengthen **A2** and **A3** in isolation, without localizing where the discussed distortions actually appear. Moreover, recent grounding-oriented works [15], [16] adopt a description-first pipeline and distill grounding annotations from existing quality descriptions [10], so their **A1** annotations are coarse, annotator-inconsistent, and introduce little new distortion evidence. As a result, grounding ends up as a downstream byproduct of description rather than its upstream evidence source, which inverts the natural distortion-to-judgment path and limits the ability of these pipelines to deliver fine-grained perception or detailed quality analysis. Additionally, some works evaluate UGC and AI-generated content (AIGC) under a shared view, yet the two differ significantly in distortion semantics: UGC images encompass diverse capturing and processing conditions and suffer from noise, blur, compression, etc., whereas AIGC prioritizes aesthetics and text-image alignment. Some UGC distortions (e.g., out-of-focus blur) may even enhance the aesthetic appeal of AIGC images. We therefore focus on UGC IQA over 10 common UGC distortions and seek a unified framework dedicated to **A1**, **A2**, and **A3** in the UGC setting.

Wenjie Liao, Jieyu Yuan, Jihong Li, Zilong Zhang and Jiachen Fu are with the VCIP, College of Computer Science, Nankai University, Tianjin 300350, China (e-mail: jasonliao@mail.nankai.edu.cn; jieyuyuan.cn@gmail.com; 120250414@mail.nankai.edu.cn; zhangzilong@mail.nankai.edu.cn; fjc@mail.nankai.edu.cn).

Yifang Xu and Haotian fan are both independent researchers (e-mail: xuy7590@gmail.com; rsgofov@outlook.com).

Chunle Guo and Chongyi Li are with the VCIP, College of Computer Science, Nankai University, Tianjin 300350, China, and also with the Nankai International Advanced Research Institute (NKIARI), Shenzhen Futian, China (e-mail: guochunle@nankai.edu.cn; lichongyi@nankai.edu.cn).

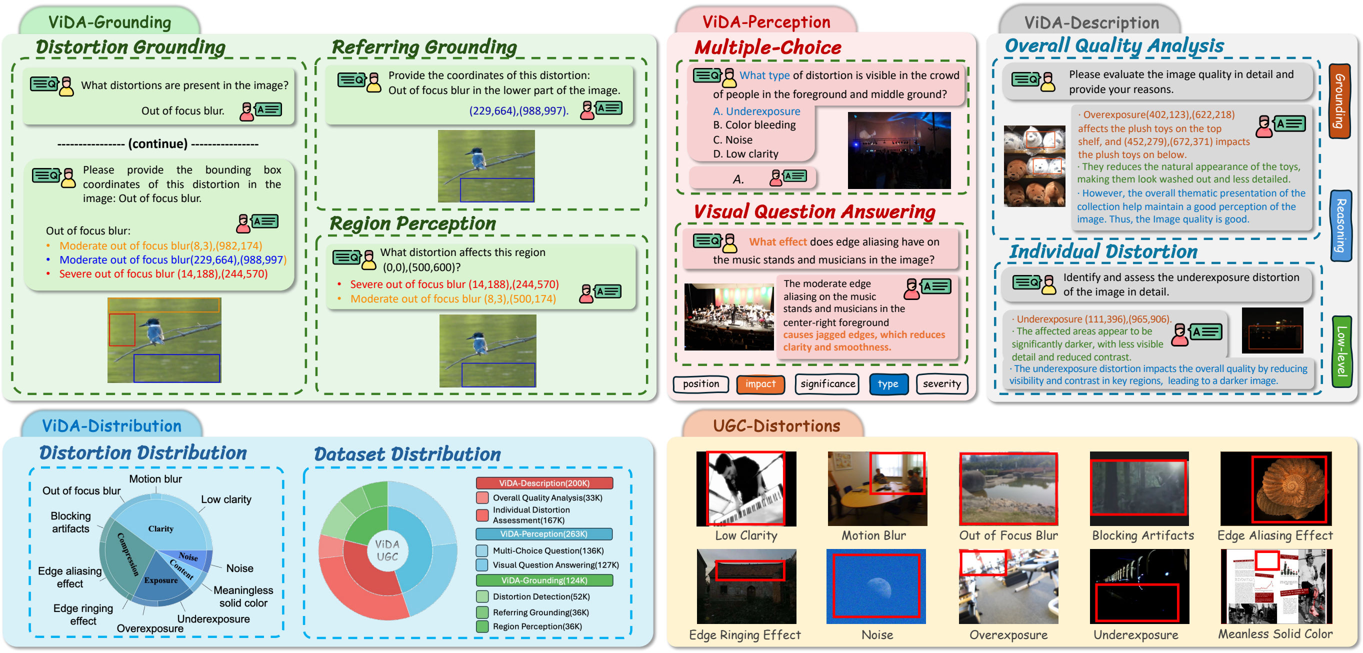

Fig. 1. An illustration of the ViDA-UGC data structure. ViDA-UGC unifies grounding, perception, and description into three sub-datasets: 1) ViDA-Grounding including distortion grounding, referring grounding, and region perception; 2) ViDA-Perception where the questions cover two formats and five distortion-oriented concerns; and 3) ViDA-Description including both overall quality analysis and individual distortion assessment. It covers 10 common UGC distortion categories and provides a large-scale instruction-tuning corpus.

To bridge **A1**, **A2**, and **A3**, we first propose **G**rounding-**P**erception-**G**uided **C**ontrolled Chain-of-Thought (**GPGC CoT**) prompting that restores the natural distortion-to-judgment path at the reasoning level. Rather than relying on open-ended CoT that may drift away from grounded visual evidence, this prompting decomposes quality analysis into progressive steps of identifying, analyzing, and reasoning about UGC distortions along an explicit distortion evidence chain, and guides intermediate steps with grounded distortion evidence. When paired with human annotations as the ground-truth evidence, it provides reliable supervision signals for dataset construction. Even without human annotations at inference time, it can guide existing MLLMs to produce more detailed and thorough quality descriptions. Building on this CoT prompting, we further propose a **distortion-oriented dataset construction pipeline**: it first samples images from UGC datasets and collects human-annotated distortion bounding boxes together with mean opinion scores (MOS), providing the evidence base for **A1**; GPT-4o [17] then generates textual distortion attributes such as position, severity, impact, and significance to enrich the low-level information required by **A2**; the GPGC CoT prompting integrates these annotations and attributes to generate reliable reasoning quality descriptions for **A3**. The raw data are converted into templated conversations for instruction tuning across the three abilities, yielding the first comprehensive **Vi**sual **D**istortion **A**ssessment Instruction Tuning dataset for **UGC** images, termed **ViDA-UGC**. ViDA-UGC consists of 11,534 images, 36K distortion bounding boxes, and 534K instruction tuning samples, organized into *ViDA-Grounding*, *ViDA-Perception*, and *ViDA-Description* sub-datasets as illustrated in Figure 1. In particular, our description data interleave textual analysis with grounding coordinates (see *ViDA-Description* examples in Figure 1), which helps link language to spatial distortion evidence and strengthens cross-modal reasoning.

To rigorously evaluate detailed IQA in the UGC setting and validate the proposed dataset, we further construct a matching benchmark. Existing explainable IQA benchmarks, such as Q-Bench [9], mainly evaluate general low-level perception and description, which is insufficient for detailed distortion assessment. To ensure comprehensive evaluation and high-quality benchmark data, we carefully select 476 images with 476 overall quality analyses from ViDA-Description, 2,567 multiple-choice questions from ViDA-Perception, and 3,106 grounding samples from ViDA-Grounding. A professional team of image-processing researchers are invited to verify the distortion evidence and reject the low-quality question-answer pairs. The resulting benchmark, termed **ViDA-UGC-Bench**, explicitly evaluates the three core abilities: for **A1**, we use distortion-centered grounding tasks and report standard localization metrics such as COCO mAP and $Acc_{0.5}$ [15], [16]; for **A2**, we adopt multiple-choice questions on distortion type, severity, position, and significance, and report accuracy; for **A3**, we evaluate overall quality analyses by GPT-based scoring on completeness, precision, and reasoning, and summarize them with a weighted final score [9]. Experiments show that current MLLMs suffer substantial performance drops on ViDA-UGC-Bench compared with Q-Bench, highlighting their limitations in detailed distortion assessment for UGC images. Meanwhile, fine-tuning on ViDA-UGC consistently improves grounding, perception, and description across Qwen-VL and InternVL series, with the best-tuned model even surpassing GPT-4o on both benchmarks. Our **contributions** can be summarized as follows:

- We introduce a distortion-oriented dataset construction pipeline combining human subject annotation, GPT-4o generation, and GPGC CoT prompting. With this pipeline, we build ViDA-UGC, the first comprehensive instruction tuning dataset for detailed IQA on UGC images through visual distortion assessment.
- We propose ViDA-UGC-Bench, which evaluates MLLMs on fine-grained distortion grounding, detailed low-level perception, and reasoning quality description capabilities in a more practical and challenging UGC setting.
- We propose Grounding-Perception-Guided Controlled CoT prompting that decomposes quality analysis into progressive distortion reasoning, guides intermediate steps with grounding and perception evidence, improves the quality-description ability of existing MLLMs, and supports higher-quality data construction for detailed IQA.

The remainder of the paper is organized as follows. Section II presents the background on UGC IQA datasets, instruction tuning, and CoT prompting. Section III presents ViDA-UGC, including the distortion-oriented data construction pipeline and GPGC CoT prompting. Section IV introduces ViDA-UGC-Bench and its evaluation settings. Section V reports the experimental results. Section VI concludes the paper.

II. Related Work

Our work aims to build a unified framework for detailed UGC IQA that jointly supports grounding, perception, and description. The unified framework is built upon a dataset, a reasoning scheme, and a benchmark. This section reviews three closely related lines of research: UGC-IQA datasets and benchmarks, instruction tuning for MLLMs, and Chain-of-Thought prompting.

A. UGC-IQA Datasets and Benchmarks.

Traditional IQA datasets mainly collect mean opinion scores (MOS) on authentically or synthetically distorted images and support score-prediction benchmarks [6], [18]–[22]. These resources are fundamental for modeling perceptual quality, but they reduce quality supervision to scalar labels and therefore provide limited support for interpretable quality analysis. With the rise of MLLMs, Q-Bench [9] introduced a benchmark for low-level perception, description, and overall quality assessment, shifting evaluation from pure score prediction toward language-based quality understanding. Subsequent works such as Q-Instruct [10], VisualCritic [13], Co-Instruct [11], DepictQA [14], and DepictQA-Wild [12] further expanded descriptive IQA and low-level instruction tuning, while grounding-oriented IQA studies such as Q-Ground [15] and Grounding-IQA [16] began to localize distortion-related evidence. However, existing resources still do not provide a unified UGC-oriented framework in which grounding, perception, and description are all organized around the same distortion annotations and evidence chain. Compared with prior public IQA resources (see Table I), ViDA-UGC jointly supports these three abilities and links them through distortion-centered supervision, making it better aligned with detailed UGC quality analysis.

B. Instruction Tuning.

Instruction tuning [23] improves the ability of the model to follow task specifications and adapt to downstream behaviors through curated instruction-response supervision. In low-level vision, targeted instruction tuning has become an effective way to strengthen MLLMs on quality-related perception and description [10]–[14]. Meanwhile, recent general-purpose MLLMs [24]–[26] already exhibit strong low-level visual skills and competitive low-level-vision performance, so the remaining bottleneck is no longer whether they can discuss low-level cues at all, but whether they can perform detailed IQA grounded in distortion evidence. What remains missing is instruction data that simultaneously teaches the model to localize distortion evidence, recognize distortion-specific attributes, and convert them into explicit quality reasoning. This gap motivates a more comprehensive distortion-oriented instruction tuning dataset for UGC images.

C. CoT Prompting.

Chain-of-Thought (CoT) prompting [27]–[29] decomposes complex tasks into intermediate reasoning steps, making model decisions more structured and interpretable. Recent frontier systems [30], [31] further demonstrate the value of stronger reasoning mechanisms, whether they are exposed explicitly or absorbed into the model. For detailed IQA, however, generic CoT is not sufficient by itself. When intermediate reasoning becomes overly diffuse or weakly tied to grounded visual evidence, the model can introduce inaccurate or unsupported observations, and the final quality analysis may become hallucinated or internally inconsistent. Our work therefore adopts Grounding-Perception-Guided Controlled CoT prompting, guiding the model to progressively identify distortions, analyze their attributes, and reason about their impact on image quality under an explicit evidence chain. This controlled prompting better matches the distortion-to-judgment reasoning process required for detailed UGC quality analysis and also supports higher-quality supervision for the description ability.

III. ViDA-UGC

This section presents the construction of **ViDA-UGC**, the first instruction-tuning dataset that jointly supports grounding, perception, and description for detailed UGC IQA. We first introduce the general principles that unify these three abilities around distortion evidence. We then describe the distortion-oriented construction pipeline, which samples representative UGC images, collects human distortion evidence, enriches distortion attributes, and organizes the resulting data into ability-aligned instruction-tuning subsets. Finally, we detail the Grounding-Perception-Guided Controlled CoT (GPGC CoT) prompting used to generate grounded reasoning supervision.

TABLE I
COMPARISON OF EXISTING PUBLIC IQA DATASETS AND THE PROPOSED VIDA-UGC. DG, RG, RP, MCQ, AND VQA RESPECTIVELY REPRESENT DISTORTION GROUNDING, REFERRING GROUNDING, REGION PERCEPTION, MULTIPLE-CHOICE QUESTION, AND VISUAL QUESTION ANSWERING.

Dataset	Quality Scoring	Quality Grounding			Quality Perception		Quality Description		
	MOS	DG	RG	RP	MCQ	VQA	low-level related	with grounding	CoT reasoning
Traditional datasets	✔	✘	✘	✘	✘	✘	✘	✘	✘
Q-Instruct-200K [10]	✔	✘	✘	✘	✔	✔	✔	✘	✘
DepictQA-Wild [12]	✔	✘	✘	✘	✘	✔	✔	✘	✔
QGround-100K [15]	✔	✔	✔	✘	✔	✔	✔	✘	✘
ViDA-UGC (ours)	✔	✔	✔	✔	✔	✔	✔	✔	✔

A. General Principles

Detailed IQA requires explaining what degrades image quality, where the relevant evidence appears, how severe the degradation is, and why it affects the final quality judgment. For user-generated content (UGC) images, this goal naturally leads to **visual distortion assessment**. Human quality analysis usually follows an interpretable path from visual distortions to quality judgment: first identify the distorted evidence, then perceive its low-level attributes, and finally organize these observations into explicit quality analysis. We therefore formulate detailed IQA for UGC around three core abilities: grounding, perception, and description.

Ability 1: Grounding (A1). Prior work in explainable IQA methods represent local quality evidence through natural language, and recent grounding-oriented studies [15], [16] have advanced this line by localizing visual cues that support quality judgment. For detailed IQA, such cues are often insufficiently explicit, as reliable quality reasoning requires spatial grounding of degradations rather than indirect associations with scene regions. Rather than grounding important objects or broad regions that affect image quality, we ground *both global and local distortion regions* directly [16]. This distinction is critical for UGC IQA, because it is the primary visual evidence of quality degradation. Explicit distortion grounding thus provides a task-aligned spatial basis for perception and description.

Ability 2: Perception (A2). In low-level visual perception, perception is commonly understood as recognizing visual attributes that determine image quality [9]. Detailed IQA extends this formulation and demands finer-grained attributes that carry diagnostic value for the final quality judgment. We accordingly redefine perception as the recognition of *distortion-specific attributes*, including type, position, severity, impact, and significance. These distortion-centered attributes constitute the elemental evidence that links grounded distortions to the subsequent quality reasoning.

Ability 3: Description (A3). Quality description has progressed from scalar scoring to generating natural-language analyses that articulate image defects and quality judgments [10]. These analyses often remain at the level of surface observations, without explicit reasoning over the underlying distortions. In detailed IQA, description should instead organize grounded distortions and perceived attributes into coherent reasoning that explains *how* and *why* the image reaches its quality judgment. We formulate the description as *distortion-grounded reasoning quality analysis* covering both overall quality assessment and individual distortion assessment. This formulation makes the final analysis traceable back to concrete distortion evidence, which aligns machine-generated descriptions more closely with how humans reason about image quality.

Overall, the three abilities are bound by a common **distortion-centered principle**: distortions serve as the unified evidence that grounding localizes, perception characterizes, and description reasons over. In contrast to existing explainable IQA resources that typically develop these abilities in isolation, the following subsections operationalize this principle through three coordinated components.

B. Distortion-oriented Dataset Construction Pipeline

Guided by the above principle, we construct ViDA-UGC through the four-stage pipeline shown in Figure 2.

Step 1: Sample Images and Collect Human Distortion Evidence. Because realistic UGC images vary substantially with capture devices, scene content, shooting conditions, post-processing pipelines, and distortion severities, detailed IQA should not be built on a narrow appearance space. We therefore sample images from a multi-source UGC pool summarized in Table II, covering in-the-wild IQA datasets, mobile-photo datasets, UGC video datasets, and other UGC data; for video data, we extract the first frame from each video. To reduce bias toward any single source or limited appearance pattern, we compute low-level features including colorfulness, sharpness, light contrast, entropy, and scene mode, and employ the improved data-shaping method based on Mixed Integer Linear Programming (MILP) [32] to select representative images from the source pool. We then conduct an in-lab subjective study on the selected images, where human subjects provide quality scores according to the five-level criteria in Figure 3 and annotate distortion bounding boxes, yielding both image-level quality labels and localized distortion evidence. The final dataset contains 11,534 images, each annotated by five human subjects. For each image, quality scores and distortion bounding boxes are collected and cleaned. Quality scores are invalidated and re-annotated if the difference between the maximum and minimum scores exceeds 1; otherwise, the MOS is computed as the average of the five scores. Bounding boxes for each distortion are split into global and local types with

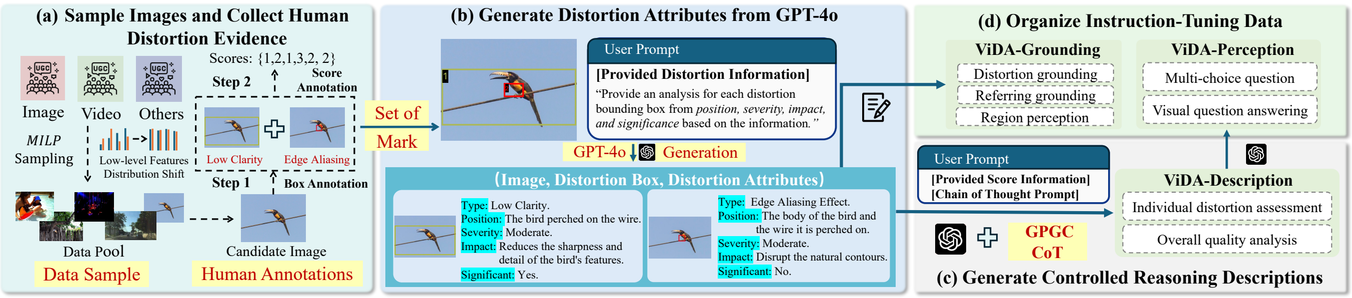

Fig. 2. Overview of the ViDA-UGC construction pipeline, which consists of four stages. (a) The MILP sampling strategy ensures an approximately uniform distribution of sampled images across low-level features, followed by human annotation of distortion regions and quality scores. (b) Each region is indexed and incorporated into GPT prompts along with IQA expertise, enabling GPT-4o to produce textual descriptions of distortions and their visual attributes. (c) These descriptions, combined with human annotations and IQA expertise, serve as ground truth in the proposed GPGC CoT prompting, where GPT-4o generates quality description data. (d) The quality descriptions are further transformed into distortion-related VQA and MCQ samples via GPT-4o, while the distortion attributes from (b) are converted into grounding data through predefined chat templates.

TABLE II
MULTI-SOURCE IMAGE POOL USED TO COVER DIVERSE UGC APPEARANCES. WE REPORT THE SOURCE CATEGORY, RAW POOL SIZE, AND THE NUMBER OF SELECTED IMAGES FROM EACH SOURCE.

Type	Source Dataset	Raw Pool Size	Selected Images
In-the-wild	KonIQ-10K [19]	10,373	1,125
	SPAQ [21]	11,125	1,146
	LIVE-FB [33]	39,810	2,039
	LIVE-itw [18]	1,169	77
Mobile Photo	DIV2K [34]	2,000	328
	DPED [35]	30,000	674
UGC Video	CVD2014 [36]	234	12
	CVQAD [37]	1,023	70
	ICME2021-UGC [38]	7,200	1,217
	Dover-UGC [39]	3,590	368
	TaoLive [40]	3,762	82
Other UGC	In-house Data	-	4,396

an area-ratio threshold of 0.7. If global boxes exist, we keep the largest global box and discard the others. Otherwise, Non-Maximum Suppression [41] is applied to suppress excessive overlap among local boxes. This process yields an average of 3.6 bounding boxes and one MOS score per image.

Step 2: Generate Distortion Attributes from GPT-4o. To keep distortion evidence as the main carrier of quality information, ViDA-UGC is centered on 10 common UGC distortions defined along 5 dimensions: *clarity, compression, exposure, content anomaly, and noise*. Previous approaches [10], [12], [13] have typically relied on either human-written quality descriptions or GPT-generated ones. However, human annotations often provide limited low-level detail, while GPT frequently produces inaccurate responses. To produce richer and more reliable distortion-centered supervision, we inject IQA expertise into GPT prompts, as shown in Figure 4(c). Specifically, we define five attributes for each distortion box: type, position, severity, impact, and significance. Based on the set-of-mark strategy [42], we overlay visual marks on the regions of distortion and provide the marked image, along with textual definitions of the distortions, to GPT-4o [17], which then generates triplets of image, distortion boxes, and distortion attributes.

Step 3: Generate Controlled Reasoning Descriptions. To ensure accurate and high-quality description data, we adopt

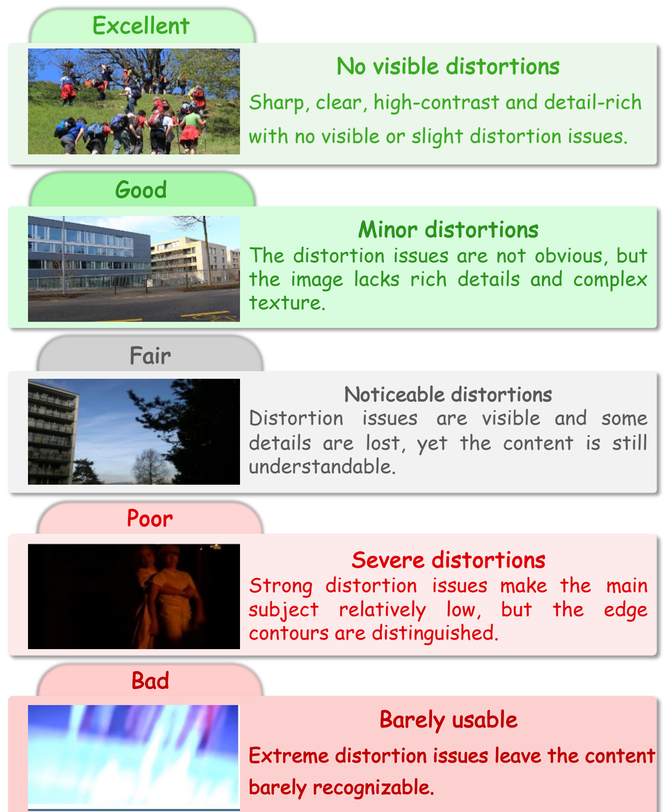

Fig. 3. Representative images corresponding to different quality levels. The five levels are further mapped to scores in $\{1, 2, 3, 4, 5\}$, where higher scores indicate better image quality.

GPGC CoT prompting and integrate MOS, rating criteria, and the corresponding distortion triplets into the GPT prompt. The detailed design of this prompting is introduced in the next subsection. Leveraging the rich low-level information derived from these inputs, GPT-4o produces logical reasoning processes for both overall quality analysis and individual distortion assessment while maintaining analytical accuracy. Finally, we interleave distortion grounding with textual descriptions in the format `[distortion](bounding box)` as *ViDA-Description* in Figure 1, which avoids mismatch between grounding outputs and quality descriptions. **Step 4: Organize Instruction-Tuning Data.** Finally, we organize the raw data into three linked instruction-tuning subsets that correspond

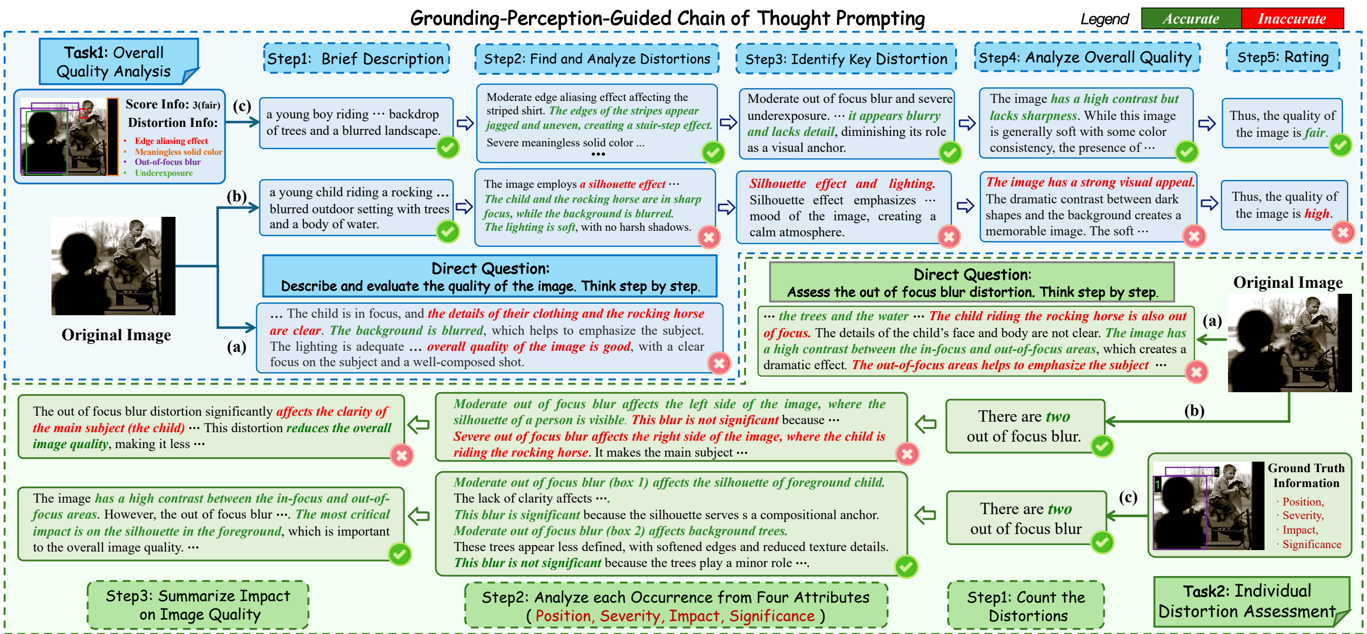

Fig. 4. Overview of our GPGC CoT prompting. The blue and green regions correspond to overall quality analysis and individual distortion assessment, respectively. (a) Direct CoT prompting produces inaccurate low-level observations and superficial reasoning. (b) Step-wise controlled CoT without explicit distortion evidence improves the reasoning structure but remains prone to hallucination and inconsistent analysis. (c) GPGC CoT incorporates human expertise and ground-truth distortion annotations to guide reliable reasoning and accurate quality prediction.

to the three detailed IQA abilities, so that the whole construction pipeline remains aligned with the distortion-centered grounding, perception, and description framework. For **A1**, we construct distortion grounding, referring grounding, and region perception through predefined chat templates. In the distortion-grounding task, the model outputs the bounding boxes of required distortions, similar to object detection. In the referring-grounding task, the model receives target distortion information and outputs the corresponding box. In region perception, the model is given a bounding box for the region of interest (RoI) and locates distortions within that RoI [43]. For **A2**, we use GPT-4o to transform quality descriptions into distortion-related visual question answering (VQA) and multiple-choice questions (MCQ). Unlike Q-Instruct, which focuses on question types such as what, how, and yes-or-no, we target distortion attributes including type, position, severity, impact, and significance, as well as other low-level features such as lighting and composition. For **A3**, the reasoning quality descriptions produced in Step 3 are directly preserved as the main supervision signal. Together, these ability-specific subsets form the three parts of ViDA-UGC shown in Figure 1.

To support **A1** with reliable human distortion evidence, the subjective study is carried out in a well-controlled laboratory environment. To ensure annotation reliability, a group of UGC-IQA experts designs the annotation protocol and training materials by drawing on professional IQA standards [44]–[47] and domain expertise. These materials include distortion definitions with representative example images and the five-level rating criteria illustrated in Figure 3. Annotators are then trained to localize distortion regions and assign quality scores on a five-point scale $\{1, 2, \ldots, 5\}$, after which they must pass a qualification exam before participating in the formal experiment. The exam consists of 300 images selected to probe two aspects of annotation competence, including image groups with identical content but different distortion types to test the annotators' discrimination ability. In addition, 30 duplicated images are inserted into the exam to test within-annotator consistency. An annotator is deemed qualified only when their labels on the duplicated images are internally consistent and agree with the ground-truth annotations.

C. Grounding-Perception-Guided Controlled CoT prompting

As detailed IQA requires explicit reasoning over distortions, we decompose the quality-description task into overall quality analysis and individual distortion assessment. We compare three prompting settings on example images from the in-lab subjective-study materials. As shown in Figure 4(a), direct CoT prompting with *"Think step by step."* often produces inaccurate low-level information and superficial reasoning. Step-wise controlled CoT in Figure 4(b) imposes a clearer reasoning structure, but it still relies on the model's implicit low-level knowledge and therefore remains vulnerable to hallucinated distortion evidence and inconsistent quality analysis. To address this limitation, our GPGC CoT prompting grounds the reasoning process in explicit distortion evidence. It uses human-annotated distortion boxes and perception attributes as guidance, so that each reasoning step follows a grounded distortion-to-judgment chain.

For overall quality analysis, the model is prompted to carry out a five-step reasoning process (Task 1 in Figure 4(c)): 1) forming a general impression of the image content; 2) identifying and analyzing distortions to gather rich low-level information; 3) isolating the key distortions that dominate the quality impression; 4) reasoning about the overall quality

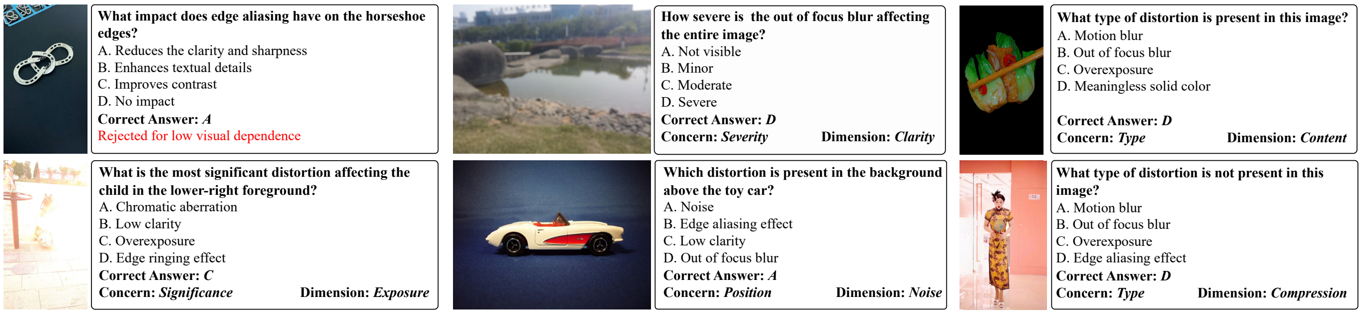

Fig. 5. Examples of selected and verified Q-A pairs: one rejected case and several accepted cases across different dimensions in ViDA-UGC-Bench.

based on the above observations; and 5) rating the image quality. The middle three steps explicitly organize distortion evidence before the final judgment, converting quality analysis from a monolithic prediction into an evidence-based reasoning process.

For individual distortion assessment, we further refine the second step into Task 2 in Figure 4(c), which requires detailed analysis of a specific distortion. This task first counts the target distortion occurrences, then analyzes each occurrence from position, severity, impact, and significance, and finally summarizes its effect on image quality. By injecting grounding and perception information into the reasoning path, GPGC CoT reduces the dependence on implicit model knowledge and produces more reliable quality descriptions.

Although GPGC CoT can guide existing MLLMs at inference time, it still depends on externally supplied distortion evidence. This observation further motivates ViDA-UGC as an end-to-end distortion-assessment dataset that teaches models to acquire grounding, perception, and description abilities jointly.

IV. ViDA-UGC-Bench

In this section, we introduce ViDA-UGC-Bench, the benchmark counterpart to ViDA-UGC. Since detailed IQA for UGC is formulated around the three core abilities of grounding, perception, and description, the benchmark is designed to evaluate these abilities under the same distortion-centered view. Starting from the collected images and data in Section III, we carefully select 476 images with their distortion triplet samples, 476 overall quality analyses from ViDA-Description, 2,567 multiple-choice questions from ViDA-Perception, and 3,106 grounding samples from ViDA-Grounding. A professional team of image-processing researchers further verifies the low-level distortion information of the question-answer pairs to reduce GPT-4o bias. Compared with existing explainable IQA benchmarks [9], [12], [48], which mostly emphasize relatively direct factual questions, our selected samples place greater emphasis on detailed distortion understanding and practical quality analysis. Examples are shown in Figure 5. The resulting benchmark covers all ten UGC distortions and provides a unified evaluation of grounding, perception, and description ability.

Benchmark on Grounding Ability. We evaluate ViDA-Grounding through three sub-tasks that test whether the model can localize distortion evidence under different conditions.

TABLE III
Completeness Evaluation Prompt of A3. [MLLM_DESC] and [DISTORTION_INFO] are replaced with the model-generated description and the reference distortion attributes in ViDA-UGC-Bench.

Completeness Evaluation Prompt
User: You are an expert in image quality assessment. Your task is to evaluate the completeness of the MLLM description based on the detailed low-level visual information derived from the reference. Evaluate whether the description [MLLM_DESC] completely includes the low-level visual information in the reference distortion information [DISTORTION_INFO]. Please rate score 3 for completely or almost completely including the reference information, 2 for including a large part of the information or a similar description, 1 for including a small part of the information or a similar description, and 0 for not including it at all. Please only provide the result in the following format: Score:

TABLE IV
Precision Evaluation Prompt of A3.

Precision Evaluation Prompt
User: You are an expert in image quality assessment. Your task is to evaluate the precision of the MLLM description [MLLM_DESC] based on the detailed low-level visual information derived from the reference distortion information [DISTORTION_INFO]. This metric penalizes low-level descriptions that significantly deviate from the reference distortion information (for example, blur for clear, significant for insignificant, colorful for monotonous, noisy for clean, and bright for dark), while minor discrepancies are not treated as violations. Please rate score 3 for no controversial low-level description, 2 for a small number of controversial low-level descriptions, 1 for a moderate number of controversial low-level descriptions, and 0 for a large number of controversial low-level descriptions. Please only provide the result in the following format: Score:

1) *Distortion grounding*: the model outputs all distortion bounding boxes in the image, and we evaluate performance using COCO mAP [49]. 2) *Referring grounding*: the model is given distortion type and location in the form "[distortion type] affecting [location]" and directly outputs the corresponding box coordinates. We evaluate this task using $\mathrm{Acc}_{0.5}$ from referring expression comprehension [50]. 3) *Region perception*: the model receives a randomly sampled region of interest and grounds distortions within that region. We again use COCO mAP as the metric.

Benchmark on Perception Ability. We adopt a multiple-choice evaluation framework following Q-Bench [9] to assess distortion-attribute perception. The benchmark focuses on four attributes that are central to our A2 formulation: type, severity, position, and significance. We exclude the impact attribute from benchmark scoring because its description is

TABLE V
REASONING EVALUATION PROMPT OF A3.

Reasoning Evaluation Prompt

User: You are an expert in image quality assessment. Please evaluate whether the reasoning in [MLLM_DESC] demonstrates comprehensive technical analysis and logical coherence.
Please rate score 3 for multi-stage reasoning with precise technical terms, score 2 for clear analysis with minor logic gaps, score 1 for basic observations with weak reasoning, and score 0 for irrelevant or illogical statements.
Please only provide the result in the following format: Score:

TABLE VI
MULTIPLE-CHOICE EVALUATION PROMPT FOR A2. "{*Question*}" AND "{*CHOICE*}" ARE REPLACED WITH THE QUESTION AND ANSWER OPTIONS DURING TRAINING AND INFERENCE.

MCQ Prompt

USER: You are an expert in image quality assessment.
`<image> <QUESTION>`
Answer with the option's letter from the given choices directly.
A. <CHOICE$_A$>
B. <CHOICE$_B$>
C. <CHOICE$_C$>
D. <CHOICE$_D$>

more subjective. Accuracy is used as the metric. Denote the image tokens as `<image>`, the question as `<QUESTION>`, and the choices as <CHOICE$_i$>; the prompt template is given in Table VI.

Benchmark on Description Ability. We evaluate ViDA-Description through *overall quality analysis*, in which the model must organize distortion evidence into coherent quality reasoning. For each image, we conduct five rounds of GPT-based evaluation between the model-generated analysis and the reference distortion information, following the protocol of Q-Bench [9], and report the averaged score to reduce the variance of GPT judgments. The evaluation covers three complementary dimensions that jointly capture descriptive quality: 1) *Completeness* measures the extent to which the model-generated analysis covers the reference distortion information; 2) *Precision* penalizes statements that conflict with the reference, such as mischaracterizing the type, position, or severity of distortions; and 3) *Reasoning* assesses whether the analysis forms a coherent evidence-to-judgment chain rather than a flat enumeration of observations. Each dimension is scored on a four-level scale {0,1,2,3}, and their weighted average serves as the final description score. The full prompt templates are listed in Tables III to V.

V. EXPERIMENT

A. Experimental Setups

We use four baseline models: Qwen-VL-Chat [3], Qwen2-VL-7B-Instruct [25], InternVL2.5-8B [24], and InternVL3-8B [26]. We evaluate their distortion grounding, low-level perception, and quality description capabilities after instruction tuning with Q-Instruct [10] and ViDA-UGC, respectively.

a) Training Details: We fine-tune all baselines with Q-Instruct and ViDA-UGC under the same task setting. Across all models, we use AdamW [51] with learning rate $1e{-}4$, weight decay 0.05, $(\beta_1, \beta_2) = (0.9, 0.95)$, a cosine learning-rate schedule with warmup ratio 0.03, ZeRO-3 [52], bfloat16 training, and low-rank adaptation (LoRA) [53] with rank 64 and alpha 128 for 3 training epochs. Qwen-VL-Chat uses 448×448 inputs with a frozen image encoder and adapter, and a batch size of $8 \times 16 \times 1$. Qwen2-VL-7B uses original-resolution inputs, LoRA on the image encoder and LLM, a fully trained adapter, merger and vision learning rates of $1e{-}5$ and $1e{-}6$, and a batch size of $16 \times 8 \times 1$. InternVL2.5-8B and InternVL3-8B use original-resolution inputs, a frozen image encoder, a fully trained adapter, and LoRA on the LLM, with a batch size of $8 \times 16 \times 1$.

b) Tasks for Evaluation: We use Q-Bench [9] to quantify general low-level visual ability. Since the LLDescribe subset in Q-Bench is not publicly available, we select 100 images from Q-Pathway in descending order of the number of human-written descriptions and remove the quality-description data from Q-Instruct to avoid test leakage. To evaluate detailed visual quality-analysis ability, we use the proposed ViDA-UGC-Bench. For distortion grounding and region perception, we additionally train three object-detection baselines [54]–[56] with distortion boxes and corresponding types so that MLLM-based grounding can be compared against dedicated detectors.

TABLE VII
REFERRING GROUNDING RESULTS FOR BASELINE AND **VIDA-UGC**-*tuned* MLLMS. RESPONSE RATE IS THE FRACTION OF VALID BOX OUTPUTS; THE BEST RESULT IS IN **BOLD** AND THE LARGEST GAIN IS IN RED.

Model	Variant	Response Rate	$Acc_{0.5}$	mIoU
Qwen-VL-Chat	Baseline	1.00	32.4	37.3
	ViDA	1.00	$41.3_{(+8.9)}$	$43.4_{(+6.1)}$
Qwen2-VL-7B	Baseline	0.79	24.9	29.8
	ViDA	0.99	$42.1_{(+17.2)}$	$45.2_{(+15.4)}$
InternVL2.5-8B	Baseline	0.98	29.0	37.0
	ViDA	1.00	$43.3_{(+14.3)}$	$46.4_{(+9.4)}$
InternVL3-8B	Baseline	0.95	25.8	33.5
	ViDA	1.00	$\mathbf{44.2}_{(+18.4)}$	$\mathbf{47.0}_{(+13.5)}$

TABLE VIII
DISTORTION GROUNDING AND REGION PERCEPTION RESULTS FOR FINE-TUNED DETECTORS AND **VIDA-UGC**-*tuned* MLLMS. DES EXTRACTS BOXES FROM QUALITY DESCRIPTIONS, DG PERFORMS DIRECT DISTORTION GROUNDING, AND RP DENOTES REGION PERCEPTION.

Type	Method	DES mAP	DG mAP	RP mAP
Detector	TOOD [54]	-	29.2	36.2
	Co-DETR [55]	-	31.4	35.8
	Grounding DINO [56]	-	37.1	42.2
MLLM	InternVL3-8B-ViDA	17.4	19.7	30.4
	InternVL2.5-8B-ViDA	15.9	20.0	32.3
	Qwen2-VL-7B-ViDA	17.9	17.7	29.4
	Qwen-VL-Chat-ViDA	15.5	16.8	27.1

B. Experiment Results

We organize the main results around the three detailed-IQA abilities: ***Grounding (A1)***, ***Perception (A2)***, and ***Description (A3)***. We first evaluate how ViDA-UGC improves **A1** relative

TABLE IX
COMPARISON OF THE **PERCEPTION** ABILITY BETWEEN BASELINE MLLMS, Q-INSTRUCT-*tuned* VERSIONS, AND **VIDA-UGC**-*tuned* VERSIONS ON Q-BENCH (LLVISIONQA-*dev*). FOR EACH BASELINE MODEL, THE HIGHEST SCORE IS HIGHLIGHTED: 1ST .

Model *(variant)*	Training Dataset	Type			Concern				*Overall*
		Yes-or-No	*What*	*How*	*Distortion*	*Other*	*I-C Distortion*	*I-C Other*	
Qwen-VL-Chat	*no* (Baseline)	56.00%	58.63%	54.77%	50.78%	61.57%	53.62%	62.45%	56.39%
	Q-Instruct	78.36%	74.56%	61.05%	70.43%	67.59%	74.34%	77.14%	71.51%
	ViDA-UGC	77.09%	78.32%	66.53%	78.60%	65.97%	79.93%	71.02%	73.98%
Qwen2-VL-7B	*no* (Baseline)	83.82%	82.74%	64.71%	75.49%	77.08%	75.66%	82.86%	77.19%
	Q-Instruct	84.00%	80.97%	65.31%	75.68%	75.69%	77.63%	80.82%	76.92%
	ViDA-UGC	82.55%	86.95%	72.62%	87.16%	70.83%	85.20%	78.37%	80.60%
InternVL2.5-8B	*no* (Baseline)	78.91%	77.21%	65.52%	69.07%	76.85%	70.07%	84.08%	73.98%
	Q-Instruct	81.64%	83.63%	64.10%	76.65%	72.00%	76.64%	83.67%	76.45%
	ViDA-UGC	81.45%	85.18%	71.60%	87.74%	66.20%	84.54%	78.37%	79.33%
InternVL3-8B	*no* (Baseline)	78.91%	76.99%	67.95%	70.43%	75.93%	71.38%	85.71%	74.72%
	Q-Instruct	76.73%	79.20%	63.69%	70.43%	70.37%	75.99%	80.41%	73.18%
	ViDA-UGC	80.00%	85.40%	66.53%	82.49%	70.37%	79.93%	74.69%	77.19%
GPT-4o	*no* (Zero-shot)	83.59%	82.40%	71.81%	75.14%	78.76%	78.10%	85.00%	78.60%

TABLE X
COMPARISON OF THE **PERCEPTION** ABILITY BETWEEN BASELINE MLLMS, Q-INSTRUCT-*tuned* VERSIONS, AND **VIDA-UGC**-*tuned* VERSIONS ON VIDA-UGC-BENCH. COMPRESS DENOTES THE COMPRESSION DIMENSION, AND CONTENT DENOTES THE CONTENT-ANOMALY DIMENSION. FOR EACH BASELINE MODEL, THE HIGHEST SCORE IS HIGHLIGHTED: 1ST .

Model *(variant)*	Training Dataset	Dimension					Concern				*Overall*
		Clarity	*Compress*	*Exposure*	*Content*	*Noise*	*Type*	*Position*	*Severity*	*Significance*	
Qwen-VL-Chat	*no* (Baseline)	41.54%	18.71%	46.09%	50.85%	13.03%	31.91%	37.83%	31.72%	36.64%	34.84%
	Q-Instruct	30.27%	39.94%	50.22%	47.46%	14.23%	28.50%	35.79%	31.72%	47.12%	35.88%
	ViDA-UGC	66.89%	69.34%	58.71%	66.10%	46.86%	60.27%	54.44%	74.37%	69.49%	63.34%
Qwen2-VL-7B	*no* (Baseline)	51.90%	52.83%	43.75%	47.46%	17.15%	43.43%	43.53%	51.26%	54.15%	47.53%
	Q-Instruct	43.17%	55.82%	39.06%	45.76%	20.92%	34.12%	42.00%	51.47%	52.08%	44.14%
	ViDA-UGC	69.45%	84.12%	69.64%	86.44%	46.86%	76.37%	61.17%	80.46%	72.20%	71.45%
InternVL2.5-8B	*no* (Baseline)	39.75%	53.77%	44.64%	55.93%	18.41%	37.08%	43.78%	44.96%	49.04%	43.51%
	Q-Instruct	34.25%	59.75%	41.96%	61.86%	32.22%	29.99%	38.32%	68.07%	45.53%	43.40%
	ViDA-UGC	73.15%	85.69%	71.43%	84.75%	60.25%	81.24%	62.06%	82.14%	78.27%	74.80%
InternVL3-8B	*no* (Baseline)	50.76%	52.52%	46.65%	55.93%	12.13%	43.57%	47.08%	47.06%	52.08%	47.37%
	Q-Instruct	31.50%	55.97%	38.84%	52.54%	28.03%	29.10%	34.90%	53.15%	47.28%	39.77%
	ViDA-UGC	71.35%	87.74%	70.31%	83.90%	46.03%	82.87%	61.80%	78.15%	72.52%	73.00%
GPT-4o	*no* (Zero-shot)	54.93%	51.26%	72.54%	58.47%	26.36%	46.38%	60.28%	53.57%	59.58%	55.20%

to generic explainable-IQA instruction tuning, establishing the local evidence needed for detailed quality analysis. We then compare results on ViDA-UGC-Bench and Q-Bench to examine how ViDA-UGC strengthens **A2** while also revealing limitations that are less visible on broader low-level benchmarks. Finally, we assess how GPGC CoT prompting further improves **A3** by guiding models to organize grounding and perception evidence into a more coherent quality judgment.

a) Grounding: For **A1**, we first evaluate how well the models can anchor detailed IQA to local distortion evidence. In the referring-grounding setting, Table VII shows that the untuned baselines perform substantially worse than on high-level grounding benchmarks [57], [58], indicating that distortion-centered localization is harder than generic object-level grounding. Fine-tuning on ViDA-UGC consistently improves both localization accuracy and output reliability. The response rate of Qwen2-VL-7B rises from 0.79 to 0.99, which is especially important because a grounding system that fails to return a valid box cannot support downstream perception or description. Table VIII further compares ViDA-UGC-tuned MLLMs with dedicated detectors on distortion grounding and region perception. Detector-based methods still achieve higher mAP on the ten-class localization task, which is expected because they are optimized for direct detection rather than open-ended analysis. However, within the MLLM family, extracting boxes from interleaved quality descriptions (DES) performs comparably to directly requesting coordinates (DG), and region perception (RP) is consistently easier than direct

TABLE XI

DESCRIPTION COMPARISON AMONG BASELINE MLLMS, Q-INSTRUCT-*tuned* VERSIONS, AND **VIDA-UGC**-*tuned* VERSIONS. RESULTS WITH * ARE OBTAINED USING THE Q-INSTRUCT PROMPT (*"Describe and evaluate the quality of the image. Think step by step."*), AND THOSE WITH † USE OUR PROPOSED GPGC COT PROMPTING. FOR EACH BASELINE MODEL, THE BEST AND SECOND-BEST SCORES ARE HIGHLIGHTED: 1ST AND 2ND.

Model *(variant)*	Training Dataset	Q-Bench				ViDA-UGC-Bench			
		Completeness	*Precision*	*Relevance*	*Overall*	*Completeness*	*Precision*	*Reasoning*	*Overall*
Qwen-VL-Chat	*no* (Baseline)*	0.72	0.53	1.86	3.11	0.20	0.58	1.56	2.34
	no (Baseline)†	0.78	0.55	1.99	3.32	0.50	0.77	2.13	3.40
	Q-Instruct*	0.97	0.76	1.95	3.68	0.58	0.93	1.98	3.49
	ViDA-UGC†	1.07	0.74	1.99	3.80	1.33	1.14	2.89	5.36
Qwen2-VL-7B	*no* (Baseline)*	1.30	1.06	2.00	4.36	0.70	0.73	2.78	4.21
	no (Baseline)†	1.36	1.28	2.00	4.64	0.74	0.92	2.88	4.54
	Q-Instruct*	1.15	1.35	2.00	4.50	0.69	1.14	1.78	3.61
	ViDA-UGC†	1.40	1.30	2.00	4.70	1.49	1.34	2.95	5.78
InternVL2.5-8B	*no* (Baseline)*	0.96	0.72	1.83	3.51	0.74	0.74	2.76	4.24
	no (Baseline)†	1.15	1.03	1.94	4.12	0.75	1.25	2.90	4.90
	Q-Instruct*	0.97	1.22	1.93	4.12	0.63	1.28	1.63	3.54
	ViDA-UGC†	1.22	1.23	1.99	4.44	1.46	1.32	2.94	5.72
InternVL3-8B	*no* (Baseline)*	1.10	0.93	1.86	3.89	0.93	0.96	2.95	4.84
	no (Baseline)†	1.27	1.34	2.00	4.61	0.96	1.28	2.98	5.22
	Q-Instruct*	0.98	1.32	1.95	4.25	0.64	1.25	1.63	3.52
	ViDA-UGC†	1.34	1.35	1.99	4.68	1.51	1.36	3.00	5.87

distortion grounding. This pattern suggests that local context helps MLLMs focus on distortion evidence even when precise coordinate prediction remains challenging. These results show that grounding is a necessary upstream ability for detailed IQA, but remains unsolved for current MLLMs.

b) Perception: For **A2**, we then compare results on Q-Bench and ViDA-UGC-Bench in Tables IX and X. This paired evaluation separates general low-level visual competence from the distortion-centered perception required by detailed IQA. Across the four untuned baselines, the overall score on ViDA-UGC-Bench is about 29% lower on average than on Q-Bench, confirming that ViDA-UGC-Bench is not a redundant re-test of Q-Bench, but a harder and more diagnostic benchmark for UGC distortion assessment. Across all four backbones, ViDA-UGC fine-tuning yields the most consistent gains. On Q-Bench, it improves the overall score for every baseline and delivers the clearest gains on distortion-related categories, showing that distortion-centered supervision transfers beyond the target benchmark. By contrast, Q-Instruct provides only modest gains for weaker models and even degrades stronger baselines such as Qwen2-VL-7B and InternVL3-8B. This contrast suggests that generic explainable-IQA supervision is insufficient once the task requires fine-grained distortion evidence. The advantage of ViDA-UGC becomes larger on ViDA-UGC-Bench, where the questions directly probe distortion type, position, severity, and significance. ViDA-UGC-tuned models consistently outperform both untuned baselines and Q-Instruct-tuned counterparts across nearly all dimensions and concerns, with especially large gains on distortion type, severity, and significance. This pattern supports the central claim of the paper: ViDA-UGC improves not only generic low-level perception, but also the reading of distortion attributes that underlie detailed IQA. Among all variants, ViDA-UGC-tuned Qwen2-VL-7B attains the best overall Q-Bench score of 80.60% and surpasses GPT-4o [17] on both Q-Bench and ViDA-UGC-Bench.

c) Description: For **A3**, we finally assess whether the models can convert distortion evidence into coherent quality analysis in Table XI. Step-wise controlled CoT without explicit distortion evidence improves description quality for untuned baselines over the generic Q-Instruct prompt, suggesting that current MLLMs contain partial low-level knowledge but express it unreliably without a structured reasoning process. Even without human annotations at inference time, GPGC CoT prompting further guides existing MLLMs to produce more detailed and thorough quality descriptions, leading to stronger reasoning and overall scores, especially on ViDA-UGC-Bench. Fine-tuning on ViDA-UGC further strengthens this ability. Relative to Q-Instruct tuning, ViDA-UGC improves not only completeness and precision, but also the reasoning score, indicating that the model is using distortion evidence more effectively rather than merely producing longer descriptions. The gains are again larger on ViDA-UGC-Bench than on Q-Bench, consistent with the distortion-centered design of both the training data and the benchmark. These results show that GPGC CoT prompting and ViDA-UGC play complementary roles: the former improves how models organize grounding and perception evidence into quality analysis, while the latter teaches models to acquire the distortion evidence needed for that reasoning.

Figure 6 provides a qualitative summary of the three abilities. Compared with GPT-4o and the Q-Instruct-tuned coun-

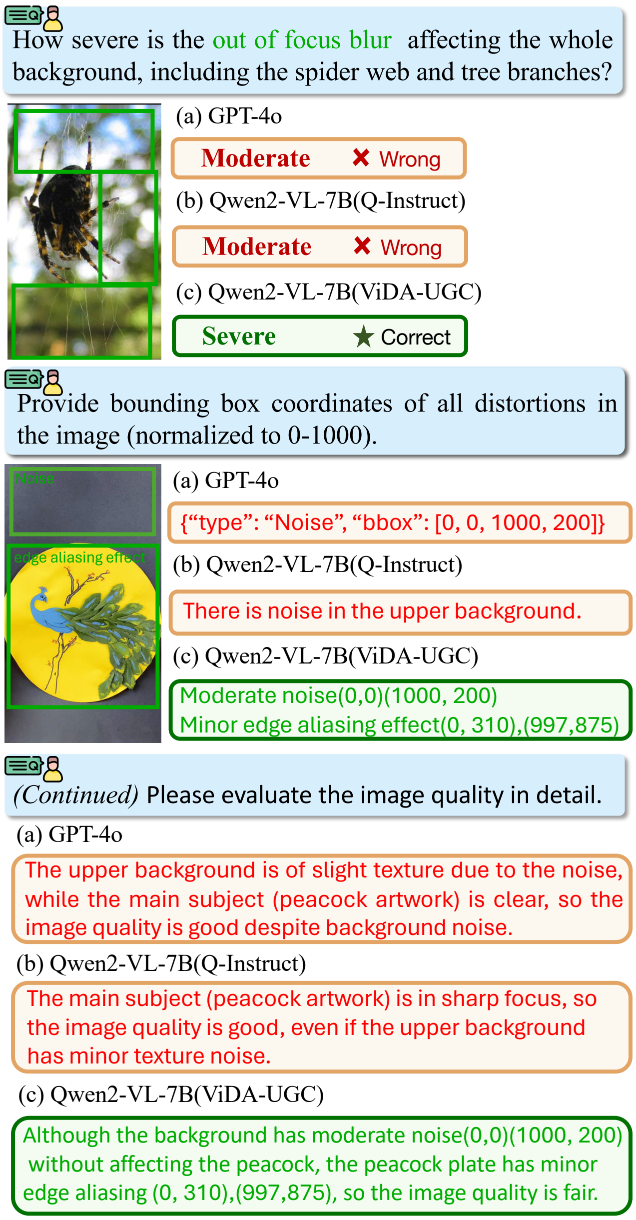

Fig. 6. Qualitative comparison across the three detailed-IQA abilities for **ViDA-UGC**-*tuned* Qwen2-VL-7B, GPT-4o, and the Q-Instruct-*tuned* counterpart. Model responses are summarized.

terpart, the ViDA-UGC-tuned Qwen2-VL-7B better localizes distortion evidence, reads distortion attributes more precisely, and ties those observations to its final quality judgment. This qualitative pattern is consistent with the quantitative results in Tables VII to XI, showing that the advantage of ViDA-UGC extends from grounding and perception to the final quality description.

VI. Conclusion

In this work, we present a unified distortion-centered framework for detailed image quality assessment of user-generated content with multimodal large language models. We formulate detailed IQA around three core abilities, namely grounding, perception, and description, and build ViDA-UGC to provide aligned supervision for these abilities through a distortion-oriented construction pipeline. Within this pipeline, we further introduce GPGC CoT prompting that organizes quality analysis along an explicit distortion-to-judgment evidence chain, enabling more informative reasoning, higher-quality description data, and more detailed quality descriptions at inference time. To evaluate this setting, we develop ViDA-UGC-Bench, a distortion-centered benchmark that provides a more diagnostic test of detailed UGC quality analysis than existing explainable-IQA benchmarks. Experimental results show that ViDA-UGC consistently improves grounding, perception, and description across multiple MLLMs, while ViDA-UGC-Bench also reveals that current models still remain limited in fine-grained distortion assessment. We hope these resources can support future progress toward more reliable and more interpretable detailed IQA.

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Wenjie Liao received the B.S. degree in Data Science and Big Data Technology from the School of Statistics and Data Science, Nankai University, Tianjin, China, in 2025. He is currently pursuing the M.S. degree in Computer Science and Technology from the School of Computer Science, Nankai University.

His research interests include computer vision, multimodal large language models, and agentic system, with a particular focus on image quality assessment.

Jieyu Yuan received the Ph.D. degree in computer technology and application from the School of Computer Science and Engineering, Faculty of Innovation Engineering, Macau University of Science and Technology, Macau, China, in 2024. He is currently a Post-Doctoral Researcher with the School of Computer Science, Nankai University, Tianjin, China.

His research interests include computer vision, 3D reconstruction, and multimodal large models, with a particular focus on underwater vision.

Yifang Xu received the M.S. degree in Control Theory and Control Engineering from the School of Logistics Engineering, Shanghai Maritime University, in 2022. She is currently an independent researcher.

Her research interests include computer vision and multimedia quality assessment, with a focus on the post-training of multimodal large language models (MLLMs).

Chunle Guo (Member, IEEE) received the Ph.D. degree from Tianjin University, China, under the supervision of Prof. Ji-Chang Guo. He was a Visiting Ph.D. Student with the School of Electronic Engineering and Computer Science, Queen Mary University of London (QMUL), U.K. He was a Research Associate with the Department of Computer Science, City University of Hong Kong (CityUHK). He was a Postdoctoral Researcher with Nankai University, under the guidance of Prof. Ming-Ming Cheng. He is currently an Associate Professor with Nankai University.

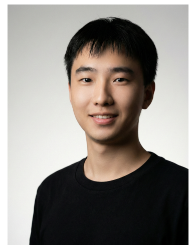
His current research interests include image processing, computer vision, and deep learning, particularly in the domains of image restoration and enhancement.

Jihong Li received the B.S. degree in computer science and technology from the School of Computer Science and Technology, Hainan University, Haikou, China, in 2025. He is currently pursuing the Ph.D. degree with the College of Elite Engineers, Nankai University, Tianjin, China.

His research interests include computer vision and image generation.

Zilong Zhang is currently a Ph.D. student at the College of Elite Engineers, Nankai University, supervised by Prof. Chongyi Li. He received a B.S. degree in Engineering from Hainan University in 2025.

His research interests focus on low-level computer vision and computational photography.



Jiachen Fu is a third-year undergraduate student at the VCIP Lab, Nankai University, under the supervision of Prof. Chongyi Li and Prof. Chun-Le Guo. He has been a member of the Excellent Class of Computer Science at Nankai University since September 2023. He is currently undertaking an Algorithm Engineer internship at ByteDance.

His research interests include Multimodal Understanding, Large Language Models, and AIGC Detection.

Haotian Fan received the B.S. degree from The Hong Kong University of Science and Technology (HKUST), Hong Kong, China, and the M.S. degree from King's College London, University of London, UK.

He is currently an independent researcher focusing on vision-language model (VLM) post-training, video generation, and the construction of multimedia quality assessment systems.

Chongyi Li (Senior Member, IEEE) received the Ph.D. degree from the School of Electrical and Information Engineering, Tianjin University, Tianjin, China, in June 2018. From 2016 to 2017, he was a Joint Training Ph.D. Student with The Australian National University, Australia. He was a Research Fellow with the City University of Hong Kong and Nanyang Technological University from 2018 to 2021. He was a Research Assistant Professor with the School of Computer Science and Engineering, Nanyang Technological University, from 2021 to 2023. He is currently a Full Professor with the School of Computer Science, Nankai University, China.

His current research interests include image processing, computer vision, and deep learning, particularly in the domains of image restoration and enhancement.